

BANA 4373 — Final Group Project Guide

Applied Data Tools for Economics and Business | Spring 2026

Dr. Fidel González | Sam Houston State University

Presentations: Tuesday, May 6, 2026 | 10:15 AM – 12:15 PM | SHB 202

How to use this guide: **Part I** covers requirements, milestones, the rubric, and tips for success. **Part II** is a complete example analytical report showing what a strong submission looks like (housing prices + FRED macro data). **Part III** is a presentation guide with a slide-by-slide outline, timing breakdown, and Q&A prep.

PART I: PROJECT HANDBOOK

1. Overview

The final group project is the capstone of BANA 4373. It is your opportunity to demonstrate the full data science workflow — from raw data to actionable insight — on a real economics or business question that your team cares about.

Weight: 30% of your final course grade **Team size:** 3–4 students **Core idea:** End-to-end pipeline on an original economics, business, or policy question.

Unlike the midterm (individual, narrower scope), the final project requires genuine team collaboration, a longer analytical report, a reproducible GitHub repository that a stranger could clone and run, and a live stakeholder-ready presentation during the final exam period.

2. What You Are Building

Your team will produce **three deliverables**, all due or presented on **May 6**:

#	Deliverable	Format	Rubric Weight
A	Reproducible repository	GitHub repo + README	Part of Pipeline & Documentation (30%)
B	Analytical report <i>or</i> Dashboard + executive brief	8–12-page PDF report, OR interactive dashboard + 2–3-page brief	Analytical Rigor (35%) + Insight/Visual (25%)
C	Stakeholder presentation	12–15 minutes live	Professionalism & Teamwork (10%)

You choose **one** report format (B): either the long-form analytical report (8–12 pages) or the dashboard + executive brief. Most teams will find the analytical report more straightforward. Choose the dashboard route only if your research question genuinely calls for it — e.g., a monitoring tool or scenario explorer.

3. Timeline and Milestones

Milestone	Due	What to Submit
Proposal	April 20	1–2 page PDF: research question, data, methods, division of labor

Milestone	Due	What to Submit
Progress Check	April 28 (in class)	Working ETL + 2 visuals + preliminary model — show on screen (3–5 min)
Final Deliverables	May 6, 10:15 AM	Repo link + report PDF submitted to Blackboard before class begins
Presentations	May 6, 10:15 AM – 12:15 PM	Live 12–15 min presentation in SHB 202

Milestone 1 — Proposal (Due April 20)

A 1–2 page PDF submitted to Blackboard. It must contain:

- **Research question(s):** What are you trying to understand or predict? Be specific.
- **Motivation:** Why does this question matter? Who is the audience?
- **Data sources:** What datasets will you use? Where will you get them?
- **Proposed methods:** OLS, DiD, ML model, etc. — and why.
- **Division of labor:** One sentence per team member listing their primary responsibilities.

■ **Tip:** A sharp research question is the most important thing in your proposal. 'What affects house prices?' is too broad. 'Do rising mortgage rates suppress home price growth more in high-cost metros than in affordable ones?' is a question your data can actually address.

Milestone 2 — Progress Check (April 28, in class)

An informal 3–5 minute screen-share in class. Come ready to show: a working ETL pipeline, at least two exploratory visualizations, and a preliminary model output (even if not final). The goal is to catch problems early — data access issues, unclear questions, scope creep — before the final week. No formal submission required.

Final Deliverables (Due May 6, 10:15 AM)

All materials uploaded to Blackboard **before** class begins. Your presentation follows during the class period.

4. Deliverable Specifications

4.1 Deliverable A — Reproducible Repository

A GitHub repository containing all code and documentation needed for a grader to reproduce every table and figure in your report from scratch. Minimum required structure:

```
your-team-repo/
  README.md           ← Required (see spec below)
  data_raw/           ← Raw files or API instructions
  data_clean/         ← Cleaned, merged datasets
  notebooks/
    01_etl.ipynb      ← Data collection and cleaning
    02_eda.ipynb      ← Exploratory analysis
    03_analysis.ipynb ← Main models and results
  exports/            ← Final figures (PNG) and tables
  requirements.txt     ← Python packages with versions
```

README.md must include:

- One-paragraph project summary (question, data, finding)

- Step-by-step instructions to run notebooks in order (assume a fresh Python environment)
- Data sources with direct links or API instructions
- Team member names and GitHub handles
- Where to put API keys (never commit actual keys to GitHub)

■ **API Keys:** *Never hard-code your actual API key in a notebook that gets pushed to GitHub. Use a placeholder: `FRED_API_KEY = "YOUR_KEY_HERE"` and note this in the README.*

4.2 Deliverable B — Analytical Report (8–12 pages)

Recommended section structure:

Section	Length	Content
Executive Summary	~0.5 page	Key question, key finding, key implication — written for a non-technical decision-maker
Research Question & Motivation	~1 page	What are you investigating? Why does it matter? Who is your audience?
Data	~1–1.5 pages	Sources, variables, time period, sample size, cleaning decisions. Include a summary stats table.
Methodology	~1–1.5 pages	Method choice and justification. Define your model equation. Justify specification decisions.
Results	~2–3 pages	Main findings with ≥ 3 figures and one results table. Walk through outputs — don't just paste them.
Limitations	~0.5–1 page	What can't your analysis do? What assumptions are you making?
Practitioner Implications	~0.5–1 page	Table or bullets matching findings to specific audiences.
Conclusion	~0.5 page	Summarize and point forward.
References	As needed	At minimum: data sources. Cite any papers or textbooks referenced.

■ See **Part II** of this guide for a complete worked example of a strong analytical report.

4.3 Deliverable B (Alternative) — Dashboard + Executive Brief

Choose this only if your project is well-suited to an interactive tool. Requirements:

- A Plotly Dash (or equivalent) dashboard with: one interactive time-series chart, one adjustable parameter (dropdown/slider), and one model output displayed.
- A 2–3 page executive brief (PDF): research question, key findings, how to use the dashboard, and limitations.
- The dashboard must be reproducible — the grader must be able to run it locally from your repo.

5. Grading Rubric

Dimension	Weight	What We're Looking For
Pipeline & Documentation	30%	ETL is clean and reproducible; README works; code is readable; notebooks run top-to-bottom; data provenance documented

Dimension	Weight	What We're Looking For
Analytical Rigor	35%	Sharp research question; appropriate methods correctly implemented; results correctly interpreted; limitations honestly assessed
Insight & Visual Communication	25%	Visualizations are clear, labeled, and informative; narrative tells a coherent story; executive summary is accessible
Professionalism & Teamwork	10%	Presentation is polished and within time; all team members contribute; Q&A; handled thoughtfully

Pipeline & Documentation (30%) — Score Guide

Score	Description
27–30	Repo clones cleanly; notebooks run top-to-bottom without errors; README is complete and accurate; all data sources documented; code is commented
21–26	Minor reproducibility issues (one easily-fixed broken step); README mostly complete; code is understandable
15–20	Significant reproducibility issues; README missing key steps; code is hard to follow
< 15	Cannot reproduce results from the repo; no meaningful README

Analytical Rigor (35%) — Score Guide

Score	Description
32–35	Sharp, addressable research question; methods are appropriate and correctly specified; coefficients correctly interpreted; assumptions stated; limitations honest and specific
25–31	Good question; methods reasonable but specification could be stronger; interpretation mostly correct
18–24	Research question is vague; methods choice not well-justified; interpretation has meaningful errors
< 18	Fundamental methodological problems; results cannot be trusted

6. Choosing a Strong Research Question

A strong research question for this project has these properties:

- **Answerable with available data.** You can actually get the data you need via FRED, BLS, Census API, Kaggle, or a company dataset.
- **Specific enough to test.** Not 'what affects GDP?' but 'does state minimum wage growth predict county-level retail employment growth in Texas?'
- **Has a clear audience.** A business analyst? A policymaker? An investor? Knowing your audience shapes the framing.
- **Uses at least two tools from this course.** ETL + regression + visualization is the baseline. DiD, A/B testing, text/sentiment, or ML methods are welcome additions.

Suggested topic areas:

Area	Example Question	Primary Method
Housing	Do rising mortgage rates predict slower price growth? Does the effect differ by metro tier?	OLS, event study

Area	Example Question	Primary Method
Labor	Did minimum wage increases in one state reduce employment vs. neighboring states?	DiD
Macro	What indicators (yield curve, unemployment, sentiment) best predict recession?	Logistic regression, ML
Retail	Does consumer sentiment predict retail sales growth?	OLS + VADER sentiment
Energy	Do oil price shocks predict CPI inflation more than wage growth does?	OLS, VAR-lite
Finance	Can macro predictors explain bank stock returns?	OLS, Ridge regression
Policy	Did ACA Medicaid expansion reduce uncompensated care in expansion states?	DiD

7. Division of Labor Recommendations

Role	Primary Responsibility
Data Lead	ETL pipeline, API calls, data cleaning, README
Analysis Lead	Main models, regression output, robustness checks
Visualization Lead	All figures, dashboard (if applicable), slide design
Report/Comm Lead	Writing the report, executive summary, presentation narrative

■ ■ Everyone should understand and be able to explain every part of the project. During Q&A, any team member may be asked about any slide.

8. Tips for Success

- **Start your ETL pipeline immediately.** Data access is the most common source of delays. Try your API calls, check for missing values, and confirm your merge logic works before Week 14.
- **Write your prior before running.** Before fitting any model, write down what sign you expect each coefficient to have and why. This is both good science and helps you catch bugs — an unexpected sign often means something went wrong in the data.
- **Interpret, don't just report.** Do not paste raw regression output. Explain in plain English what each coefficient means for a practitioner.
- **Be honest about limitations.** A paper that identifies its own limitations is more credible than one that ignores them. You are not expected to solve the identification problem — just be clear about what your analysis can and cannot claim.
- **Make your visualizations speak.** Every figure should have: a descriptive title, labeled axes with units, a data source note, and enough visual contrast to be readable in a PDF printout.
- **Rehearse your presentation.** Run through it at least twice with a timer. 12 minutes goes faster than you think.

9. Common Mistakes and FAQ

Q: Can we use a dataset from Kaggle?

Yes, but document the source clearly, understand what the variables mean, and ideally supplement it with at least one API-pulled source to demonstrate that skill.

Q: Do we need to use machine learning?

No — the syllabus says 'ML use is optional but welcome.' A rigorous OLS or DiD analysis is entirely sufficient. Only add ML if it genuinely answers your question better than regression.

Q: How many visualizations do we need?

The rubric does not specify a count, but a typical strong report has 4–6 figures: a time-series overview, a scatter/correlation plot, main results visualization, and at least one diagnostic or robustness check.

Q: What counts as a 'reproducible' repo?

Clone your own repo to a fresh folder, follow your own README from top to bottom, and verify every notebook runs without modification and produces the same outputs.

Q: Can we use AI tools to help write the report?

Review the AI policy in the syllabus. AI may assist with debugging, syntax, and exploring approaches — but the analysis, interpretation, and written argument must be your own. Disclose AI assistance in an appendix note.

Q: What if team members contribute unequally?

Contact the instructor before the final week. Do not wait until after grades are posted.

Q: Can our presentation go over 15 minutes?

No. You will be stopped at 15 minutes. See Part III for a timing guide.

PART II: EXAMPLE ANALYTICAL REPORT

What Predicts U.S. Housing Price Growth? A Multi-Factor Macroeconomic Analysis Using FRED Data (2000–2023)

Team: Jane Smith, Carlos Rivera, Priya Nair, David Chen

BANA 4373 — Applied Data Tools for Economics & Business | May 6, 2026

Executive Summary

This report investigates which macroeconomic factors best predict U.S. national home price appreciation, using monthly data from the Federal Reserve Economic Database (FRED) covering January 2000 through December 2023. We estimate an OLS regression of year-over-year Case-Shiller Home Price Index (HPI) growth on four predictors: the 30-year fixed mortgage rate, the unemployment rate, housing starts, and year-over-year CPI inflation.

Key findings: Mortgage rates and unemployment are the strongest predictors. A one-percentage-point increase in the 30-year mortgage rate is associated with approximately **2.1 percentage points lower** year-over-year HPI growth. However, the post-2020 period represents a structural break: the pandemic-era combination of record-low rates, supply disruptions, and demand shifts produced a home price surge that the baseline model substantially underpredicts. An optional Ridge regression confirms that OLS findings are not artifacts of multicollinearity.

Implication: Mortgage rates are a reliable leading indicator for housing market direction, but practitioners should treat quantitative forecasts with caution given documented structural instability in the relationship.

1. Research Question and Motivation

U.S. home prices appreciated at an extraordinary pace from 2020 to 2022 — the Case-Shiller National HPI rose more than 40% over just 24 months — before cooling sharply as the Federal Reserve raised interest rates. For real estate investors, homebuilders, financial analysts, and policy researchers, understanding what drives this volatility is both practically urgent and analytically complex.

Research question: *Which macroeconomic indicators best predict year-over-year growth in U.S. national home prices — and has this relationship changed since the COVID-19 pandemic?*

This question is relevant to multiple audiences: a real estate portfolio manager deciding on asset allocation, a homebuilder timing new construction investment, or a policymaker evaluating the housing affordability implications of Federal Reserve rate decisions. We approach this as a predictive modeling question — we are not claiming causal identification — but we use economic theory to guide variable selection and interpret our coefficients with appropriate caution.

2. Data

All data were pulled from the **FRED API** (Federal Reserve Bank of St. Louis) at monthly frequency using the `fredapi` Python library.

Variable	FRED Series	Description
HPI Growth (outcome)	CSUSHPISA	S&P;/Case-Shiller U.S. National HPI, YoY % change
Mortgage Rate	MORTGAGE30US	30-year fixed mortgage rate (%)

Variable	FRED Series	Description
Unemployment Rate	UNRATE	Seasonally adjusted U-3 unemployment rate (%)
Housing Starts	HOUST	Total privately owned housing starts (thousands, SAAR)
CPI Inflation	CPIAUCSL	CPI all urban consumers, YoY % change

Sample period: January 2000 – December 2023 (N = 287 monthly observations after computing year-over-year changes and aligning series).

Cleaning decisions: The Case-Shiller HPI is reported with a two-month publication lag; we align all series to the same calendar month. April 2020's extreme unemployment spike (14.7%) is retained in the main specification but examined separately in robustness checks.

Summary Statistics (Full Sample, 2000–2023):

Variable	Mean	Std Dev	Min	Max
HPI Growth (% YoY)	4.3	7.8	–18.9	20.1
Mortgage Rate (%)	5.4	1.6	2.7	8.5
Unemployment Rate (%)	5.8	2.0	3.4	14.7
Housing Starts (000s)	1,247	365	470	2,273
CPI Inflation (% YoY)	2.4	1.9	–2.1	9.1

3. Methodology

3.1 Baseline OLS Model

We estimate the following linear regression:

$$\text{HPI_Growth}(t) = \beta_0 + \beta_1 \times \text{MortgageRate}(t) + \beta_2 \times \text{Unemployment}(t) + \beta_3 \times \text{HousingStarts}(t) + \beta_4 \times \text{CPI_Inflation}(t) + \varepsilon(t)$$

Variable selection rationale:

Variable	Expected Sign	Economic Rationale
Mortgage Rate	$\beta \leq 0$	Higher rates reduce affordability and demand, slowing price appreciation
Unemployment Rate	$\beta \leq 0$	Higher unemployment reduces income and confidence, dampening demand
Housing Starts	$\beta \leq 0$	More new construction adds inventory, putting downward pressure on prices
CPI Inflation	$\beta \geq 0$	Higher general inflation raises nominal prices; real estate is an inflation hedge

Standard errors: Newey-West (HAC) with a 12-month lag window corrects for autocorrelation in monthly time-series data. Standard OLS standard errors would understate uncertainty in this setting.

3.2 Post-2020 Structural Break Test

We estimate the model on two sub-samples: (i) January 2000 – February 2020 and (ii) March 2020 – December 2023. Comparing coefficient estimates across sub-samples reveals whether the pandemic fundamentally changed the predictive relationship.

3.3 Ridge Regression Extension (Optional ML)

As a robustness check, we fit a Ridge regression (L2 regularization) using 5-fold cross-validation to select the penalty parameter λ . If Ridge coefficients are qualitatively similar to OLS, this supports the stability of our findings and confirms that mild multicollinearity is not distorting results.

4. Results

4.1 Exploratory Analysis

Figure 1 (Time Series Overview) shows all five variables over 2000–2023. Four distinct housing market regimes are visible:

- **2000–2006:** Classic bubble. HPI growth accelerated to 15–18% YoY while mortgage rates were moderate (5.5–6.5%) and unemployment was low. The divergence of price growth from fundamentals is evident.
- **2007–2012:** Bust and slow recovery. HPI growth collapsed to –18.9% at the trough. Unemployment spiked above 10%. Mortgage rates fell but demand destruction overwhelmed any affordability benefit.
- **2013–2019:** Steady recovery. HPI growth of 4–7% YoY, unemployment declining toward 3.5%, housing starts recovering toward historical norms.
- **2020–2023:** Pandemic shock. Record-low mortgage rates (2.7% in January 2021) combined with a demand surge drove HPI growth to 20.1% at the peak. The subsequent rapid rate hike cycle (3% → 7%+ in 18 months) produced the sharpest slowdown since the financial crisis.

Figure 2 (Scatter Matrix) confirms the expected negative correlations: mortgage rates and HPI growth ($r \approx -0.41$), unemployment and HPI growth ($r \approx -0.38$). Post-2020 observations (highlighted) show notably more scatter — prices initially surged despite rising rates, reflecting demand-side forces temporarily overwhelming affordability constraints.

Figure 3 (Rolling 36-Month Correlations) shows that the mortgage rate–HPI correlation is not stable over time. It strengthens in 2004–2009 and again in 2018–2023, but weakens during 2010–2017, suggesting the relationship is episodic rather than constant — an important limitation for any out-of-sample forecasting.

4.2 Regression Results

Table 1: OLS Regression Results (Newey-West HAC standard errors, N=287)

Variable	Exp. Sign	Coefficient	Std Error	t-stat	Significant?
Intercept	—	18.4	4.2	4.38	Yes (1%)
Mortgage Rate (%)	Negative	–2.08	0.61	–3.41	Yes (1%)
Unemployment Rate (%)	Negative	–1.24	0.38	–3.26	Yes (1%)
Housing Starts (000s)	Negative	–0.003	0.003	–1.00	No (n.s.)
CPI Inflation (% YoY)	Positive	+1.41	0.42	3.36	Yes (1%)
R-squared	—	0.41	—	—	—

Key coefficient interpretations:

- **Mortgage Rate (−2.08):** Holding other variables constant, a one-percentage-point increase in the 30-year mortgage rate is associated with approximately **2.1 percentage points lower** year-over-year HPI growth. The difference between a 3% and a 7% rate implies roughly 8 pp of expected HPI growth in either direction — consistent with the observed 2020–2023 cycle.
- **Unemployment Rate (−1.24):** A one-point increase in unemployment is associated with approximately **1.2 percentage points lower** HPI growth. The COVID unemployment spike (3.5% → 14.7%) would, all else equal, predict a 13.8 pp decline in HPI. The fact that we observed the opposite reflects the extraordinary offsetting effect of record-low mortgage rates and fiscal transfers.
- **Housing Starts (−0.003, n.s.):** Not statistically distinguishable from zero. Supply effects may operate on longer time horizons than our monthly specification captures, or may be too local to appear in national aggregates.
- **CPI Inflation (+1.41):** As expected, higher general inflation is associated with higher nominal home price appreciation — consistent with real estate's role as an inflation hedge.

R² = 0.41: The model explains about 41% of the variance in monthly HPI growth — meaningfully better than a simple two-variable model, but leaving substantial unexplained variation driven by local market dynamics, institutional demand, and structural shifts the model cannot capture.

4.3 Sub-Sample Analysis

Table 2: Coefficient Stability Across Sub-Samples

Variable	2000 – Feb 2020	Mar 2020 – Dec 2023
Mortgage Rate	−1.71**	−3.82**
Unemployment Rate	−1.18**	−0.43 (n.s.)
CPI Inflation	+0.92**	+2.65**
R-squared	0.39	0.61

The mortgage rate coefficient is substantially larger in absolute value in the post-2020 period — the sensitivity of home prices to rates roughly doubled. The unemployment coefficient becomes statistically insignificant post-pandemic — consistent with fiscal transfers and remote-work demand shifts decoupling price growth from labor market conditions. This is a meaningful structural finding, not just a parameter shift.

Figure 4 (Residual Plot) confirms this: the model's largest errors cluster in March–December 2020 (prices rose more than predicted: demand surge) and March–October 2022 (prices fell faster than predicted: rate shock). These are regime-change episodes that a regression trained on prior decades cannot anticipate.

4.4 Ridge Regression Robustness Check

Variable	OLS Coefficient	Ridge Coefficient ($\lambda=0.8$)
Mortgage Rate	−2.08	−1.94
Unemployment Rate	−1.24	−1.15
Housing Starts	−0.003	−0.004
CPI Inflation	+1.41	+1.38

Ridge estimates are qualitatively identical to OLS with minor shrinkage. Multicollinearity is not materially distorting results. The predictors are correlated (mortgage rates and unemployment have $r = 0.28$) but not severely enough to destabilize OLS coefficient estimates.

5. Limitations

- **Association, not causation.** Our regression identifies statistical associations. Mortgage rates respond to the same macro conditions that drive housing demand. A genuine causal estimate would require an instrumental variables strategy — e.g., using shifts in Treasury yields driven by foreign capital flows as an instrument for domestic mortgage rates.
- **Structural instability.** The sub-sample analysis confirms that pre-2020 parameter estimates do not transfer cleanly to the pandemic period. Any practitioner using this model for out-of-sample forecasting should treat post-2020 estimates as provisional.
- **National aggregation.** Housing markets are intensely local. A national regression smooths over enormous geographic variation. City- or metro-level panel analysis would be more policy-relevant.
- **Omitted variables.** Investor activity, demographic trends (millennial peak household formation), zoning constraints, and migration patterns all plausibly affect home prices and are absent from our model.
- **Lagged effects.** We use contemporaneous predictors. In reality, housing market decisions respond to rates with a lag (mortgage rate locks, construction timelines). A distributed lag specification may improve fit and economic plausibility.

6. Implications for Practitioners

Audience	Key Takeaway
Real estate investor	Mortgage rate movements are your most important macro signal. A 100 bps rise in the 30-year rate is historically associated with ~2 pp of slower appreciation; a 400 bps sustained increase should be expected to produce a meaningful price correction.
Homebuilder / developer	Mortgage rate sensitivity has strengthened post-pandemic. Time new starts decisions to rate cycles with more lead time than historical data suggests.
Financial analyst / fixed income	R^2 of 41% leaves significant unexplained variance. Supplement macro indicators with local inventory, days-on-market, and pending home sales data for forecasting.
Policy researcher	The decoupling of HPI growth from unemployment in 2020–2022 points to demand-side structural shifts that traditional macro models miss. Future analysis should incorporate institutional demand and zoning data.

7. Conclusion

A four-variable OLS model using mortgage rates, unemployment, housing starts, and CPI inflation explains approximately 41% of the variance in U.S. monthly home price growth from 2000 to 2023. Mortgage rates and unemployment are statistically and economically the most significant predictors. However, a structural break analysis reveals that pre-2020 parameter estimates do not transfer cleanly to the pandemic period, during which mortgage rate sensitivity intensified and unemployment's predictive power collapsed.

The practical implication: mortgage rate movements are the single most useful macro signal for housing market direction — but any quantitative estimate of the effect should be treated as time-varying rather than fixed. An improved model would incorporate local market data, lagged predictors, and structural controls for supply elasticity.

References

- Board of Governors of the Federal Reserve System. *FRED Economic Data*. Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/>

- Case, K. E., & Shiller, R. J. (1989). The efficiency of the market for single-family homes. *American Economic Review*, 79(1), 125–137.
- Glaeser, E. L., & Gyourko, J. (2018). The economic implications of housing supply. *Journal of Economic Perspectives*, 32(1), 3–30.
- Shiller, R. J. (2015). *Irrational Exuberance* (3rd ed.). Princeton University Press.
- Stock, J. H., & Watson, M. W. (2019). *Introduction to Econometrics* (4th ed.). Pearson.

This report was produced as a group final project for BANA 4373, Spring 2026. All code is reproducible in the accompanying GitHub repository. AI tool usage: GitHub Copilot was used for autocomplete assistance in notebook cells. All analysis design, interpretation, and written argument are the authors' own.

PART III: PRESENTATION GUIDE

Presentation Overview

Item	Detail
Date	Tuesday, May 6, 2026
Time	10:15 AM – 12:15 PM
Location	SHB 202 (your regular classroom)
Length	12–15 minutes per team + 2–3 minutes Q&A;
Audience	Stakeholder-ready — lead with the question and the answer, not the methodology

■ ■ At 15 minutes you will be stopped. Plan for 13 minutes of content and 2 minutes of buffer. Rehearse with a timer — out loud.

Slide-by-Slide Outline

Slide 1 — Title Slide

■ Target: 0:30

- Project title (descriptive, not generic — e.g., 'Mortgage Rates and Home Prices: A Macro Analysis')
- Team member names, course, date
- One compelling chart used as a visual hook — the audience should be curious before you speak

Slide 2 — The Hook: Why Should Anyone Care?

■ Target: 1:00

- Open with a striking data fact or question — not 'Today we will present our project on...'
- Example: 'Between January 2021 and October 2023, the 30-year mortgage rate went from 2.7% to 7.8% — the fastest increase in 40 years. What actually drives the relationship between rates and home prices?'
- One striking fact + research question. No more.

Slide 3 — Research Question and Data

■ Target: 1:30

- State the research question in one bolded sentence
- Three bullets: data sources, time period, unit of observation
- Grader check: Is the question specific and answerable? Does the team understand their data?

Slide 4 — Key Variables: What We're Measuring

■ Target: 1:00

- Clean table: variable name | source | expected direction (+/–) | one-line economic rationale
- Do NOT describe your ETL pipeline here — that belongs in the repo
- This slide shows you understand WHY you chose these variables

Slide 5 — What the Data Looks Like (EDA)

■ Target: 1:30

- Your best time-series overview figure — one large, well-labeled chart
- Walk the audience through major episodes in the data; point to specific features
- Do NOT use six small charts on one slide — one large chart communicates more

Slide 6 — Our Approach (Methodology)

■ Target: 1:00

- Write your model equation on the slide — one line, readable by a non-technical audience
- Brief bullets on why each variable is included (one line each)
- One line on standard error choice — don't belabor technical details

Slide 7 — Main Results ★ MOST IMPORTANT SLIDE ★

■ Target: 2:00

- Spend the full 2 minutes here — this is the heart of your presentation
- A coefficient plot (visual) is more effective than a raw regression table
- For each significant predictor: state the coefficient, state what it means in plain English, confirm the sign matches your expectation
- Example: 'A one-point increase in the 30-year mortgage rate is associated with 2.1 pp slower home price growth — that's the size of effect that matters to a portfolio manager.'

Slide 8 — What the Model Gets Right and Wrong

■ Target: 1:30

- Show your residual plot or fitted vs. actual
- Highlight large errors and explain WHY the model struggles there — this shows economic understanding
- Do not apologize for the model's limitations: 'These errors tell us something important about what the model cannot capture.'

Slide 9 — Robustness Check (Optional)

■ Target: 0:30

- One slide summarizing your sub-sample analysis, Ridge regression, or sensitivity test
- Key takeaway: 'Main coefficients hold up, with one important exception: post-2020 mortgage rate sensitivity is stronger.'
- Skip this slide if you are running short on time

Slide 10 — Implications for Practitioners

■ Target: 1:00

- Two-column table: Audience | Key Takeaway
- Keep concise — the audience should leave knowing exactly what THEY would do differently with your findings

Slide 11 — Limitations and Next Steps

■ Target: 0:30

- Three bullets maximum — be specific
- 'National data masks local variation' is better than 'there are some limitations'
- Suggest one concrete next step for future research

Slide 12 — Conclusion

■ Target: 0:30

- One sentence answer to your research question. One sentence on what it means. Done.
- Do NOT summarize every slide. Do NOT say 'In conclusion, we found that...'
- Just answer the question you asked on Slide 2

Slide 13 — Thank You / Questions

■ Target: hold

- Team names, repo link or QR code, contact info
- This slide stays visible during the entire Q&A; period

12–15 Minute Timing Summary

Slide(s)	Content	Target	Cumulative
1	Title	0:30	0:30
2	Hook + Research Question	1:00	1:30
3	Data Overview	1:30	3:00
4	Key Variables	1:00	4:00
5	EDA + Time Series	1:30	5:30
6	Methodology	1:00	6:30
7 ★	Main Results	2:00	8:30
8	Residuals / Model Fit	1:30	10:00
9	Robustness (optional)	0:30	10:30
10	Practitioner Implications	1:00	11:30
11	Limitations	0:30	12:00
12	Conclusion	0:30	12:30
13	Q&A;	held	≤ 15:00

Slide Design Principles

- **Less is more.** Each slide should have one main idea. If you cannot state it in six words, the slide is doing too much.
- **Figures over bullets.** A well-labeled time-series plot communicates more than five bullet points. Use slides as a visual aid, not a teleprompter.
- **Consistent formatting.** One font family, two or three colors, consistent label sizes. No clip art, excessive animation, or decorative elements.
- **Readable from the back of the room.** Body text minimum 20pt. Chart axis labels minimum 14pt. Every figure must be legible without squinting.
- **Cite your sources inline.** Each chart should have a small source note (e.g., Source: FRED, Federal Reserve Bank of St. Louis).

Delivery Tips

- **Rehearse with a timer — out loud, not in your head.** Reading silently is 30–40% faster than speaking to an audience. What feels like 10 minutes may run 14 when delivered.
- **Divide speaking roles clearly before the day.** Each team member should know exactly which slides they own. Handoffs should be natural: 'Carlos is going to walk you through the regression results.'
- **Make eye contact with the room, not the screen.** Know your slides well enough to glance briefly and return to the audience.
- **Speak to the data, not about the data.** Don't say 'In this figure we can see that...' Say 'Home prices peaked at 20% annual growth in early 2022. The mortgage rate was 4% and rising fast — that crossover is what you're seeing on the left.'
- **Lead with findings.** Business audiences want the answer first, then the evidence. Open each analytical slide with your conclusion, then show the supporting data.

Handling Q&A;

You will have 2–3 minutes of Q&A; immediately after your presentation. Common questions:

"Why didn't you control for X?"

Acknowledge the limitation, explain briefly why it was outside your scope, and state what direction you'd expect the omitted variable bias to run. 'That's a fair point — we didn't include inventory data; its omission may cause us to overestimate the mortgage rate effect somewhat.'

"Can you explain the coefficient interpretation again?"

Have a plain-English version ready for every significant coefficient. Practice explaining it as if to a friend who hasn't taken this course.

"How confident are you in these results?"

Refer to your HAC standard errors, R^2 , and robustness checks. Be honest: 'We're confident in the direction of the mortgage rate effect — it's significant across sub-samples and robust to Ridge regularization. We're less confident in the precise post-2020 magnitude.'

"What would you do differently with more time?"

Have one concrete answer ready. Example: 'We'd run this at the metro level using FHFA HPI data to capture geographic heterogeneity, and we'd try lagged predictors to better capture the timing of rate effects.'

If you don't know the answer: 'That's a great question — I'm not certain, but my best guess is X. I'd want to verify that in the data before committing to an answer.' This is always an acceptable and professional response.

Final Checklist Before May 6

- All notebooks run top-to-bottom on a clean clone of the repo
- README is complete and accurate
- No raw API keys committed to GitHub
- Report is 8–12 pages and PDF-formatted, submitted to Blackboard
- Presentation slides are finalized and backed up (email to yourself + Blackboard upload)
- Each team member knows exactly which slides they own
- Presentation timed at 12–13 minutes in a full rehearsal

- All figures are readable: large font, labeled axes, source notes
- You can explain every significant coefficient in plain English
- You have one concrete answer prepared for 'what would you do differently?'

Good luck — make it count.

BANA 4373 | Spring 2026 | Dr. Fidel González | Sam Houston State University | Final Group Project Guide